

Student 3 - Implementation (Part 1 & 2)

Time: 2 minutes 30 seconds

Slides: Slide 05 (Implementation Part 1) + Slide 06 (Implementation Part 2)

"Now I will explain the implementation. We have six blocks across two slides."

| Click to slide 05 - Implementation Part 1.

"Block 1 is the Data model. The Item class has fields id, type, name, category, date, location, contact, status."

| Point at block 1 heading and code.

"The diagram shows 8 fields separated by pipes. id pipe type pipe name and so on."

| Point at the diagram on the right side of block 1.

"Block 2 is File storage. Each line in items.txt is one item."

| Point at block 2 heading.

"The code shows we split the line by pipe character using line dot split. Then we parse the id using Integer dot parseInt."

| Point at the code lines in block 2.

"The diagram shows the split result - 8 fields from p0 to p7."

| Point at the diagram showing p0 p1 p2 etc.

"Block 3 is Validation. We use regex to check date format."

| Point at block 3 heading.

"The regex is backslash d 2 hyphen backslash d 2 hyphen backslash d 4. This means 2 digits, hyphen, 2 digits, hyphen, 4 digits."

| Point at the regex in the code and diagram.

"15-09-2026 is accepted in green. 9/15/26 is rejected in red."

| Point at the green OK example and red REJECT example.

"Now part 2 - Search, Claim, and Menu loop."

| Click to slide 06 - Implementation Part 2.

"Block 4 is Search. We use Java streams. items dot stream dot filter dot collect."

| Point at block 4 code.

"The filter checks if the name contains the keyword. The execution flow shows items stream, filter, collect, then a filtered list."

| Point at the execution flow diagram showing the pipeline.

"Block 5 is Claim. We loop through items, find the matching id, set status to RESOLVED, then rewrite the file."

| Point at block 5 code.

"The execution flow shows before - id 7 status OPEN. After claim - id 7 status RESOLVED. Then rewrite items dot txt."

| Point at the before and after diagram in block 5.

"Block 6 is the Menu loop. We use while true to keep the program running."

| Point at block 6 code.

"The code reads user input, checks if it equals 1 to post lost, or 8 to break and exit."

| Point at the if statements in block 6.

"The execution flow shows the menu options - 1 Post Lost, 2 Post Found, 3 View Lost, 5 Search, 6 Claim, 8 Exit."

| Point at the menu list in the execution flow.

"I will now hand over for the live demonstration."

| Gesture towards next presenter.